



CS 540 Module Seven Activity Guidelines and Rubric

Overview

Professionals in data-driven fields must not only interpret data but also gather, analyze, and present it in meaningful and actionable ways. This activity allows you to practice these essential skills by working through the complete exploratory data analysis (EDA) process.

This hands-on activity mirrors real-world tasks performed by analysts, researchers, and strategists across various industries. By completing this activity, you will improve your confidence in selecting appropriate data visualization tools, supporting your analysis with evidence, and effectively communicating your findings.

Scenario

You are a data consultant for a nonprofit organization focused on improving public transportation. The organization is evaluating transit usage across three neighborhoods and is interested in identifying key patterns or disparities. Your task as a data consultant is to:

1. Gather relevant data related to public transit usage.
2. Analyze the data using Python-based visualizations.
3. Summarize your findings in a short written report to share with stakeholders.

Directions

In this activity, you will identify relevant data to generate visualizations and interpret insights. When you are ready to begin your work, access Codio by clicking the title of the next task.

When the assignment is complete, download the HTML version of your project by clicking on **File**, then **Download As**, then **HTML (.html)** from the Jupyter menu bar. Save the file to your local machine.

***Note:** if the code block was not run in the Notebook, it may not render in the HTML output of the Notebook. To fix missing blocks, run the code block, then download as HTML.*

Part One: Python Code

In this part, you will write Python code to analyze the data set provided in Codio and generate visualizations that explore trends.

1. Create multiple, distinct **visualizations** that explore different aspects of the data using Python Libraries.
 - A. Include code comments explaining each step of your analysis.

Part Two: Report

2. Explain the **trends, differences, or outliers** you observed in the data.
3. Describe any **disparities or patterns** that may be relevant to the nonprofit's goals.

- 4. Write a **summary** of what the data suggests about transit usage across neighborhoods.
- 5. Explain how the **visualizations support** your conclusions.
- 6. Discuss how your **findings** could inform the nonprofit's decision making or strategy.
- 7. Describe the importance of **data-driven insights** in addressing real-world challenges.

What to Submit

Python Code

Download your Jupyter Notebook as an HTML file. Review the file to ensure all code and visualizations are displayed.

Report

Submit your report as a 2- to 3-page Microsoft Word document with double-spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module Seven Activity Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Creates multiple, distinct visualizations that explore different aspects of the data using Python Libraries, including code comments explaining each step of the analysis	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5
Trends, Differences, or Outliers	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains the trends, differences, or outliers observed in the data	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Disparities or Patterns	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes any disparities or patterns that may be relevant to the nonprofit's goals	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5
Summary	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Writes a summary of what the data suggests about transit usage across neighborhoods	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5
Visualizations Support	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains how visualizations support the conclusions	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5
Findings	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Discusses how the findings could inform the nonprofit's decision making or strategy	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	13.5
Data-Driven Insights	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes the importance of data-driven insights in addressing real-world challenges		Does not attempt criterion	13.5
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5.5
Total:					100%